

# Final Report

RBE 502 - Robot Control

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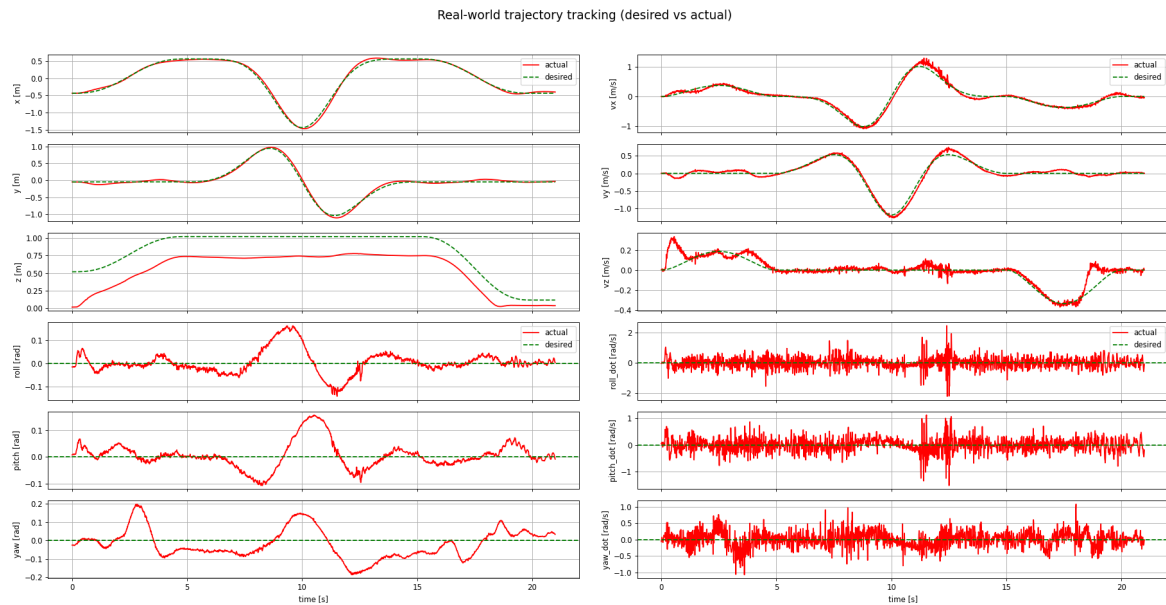
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This report details the performance evaluation of a quadrotor controller designed to execute a circular trajectory, comparing results obtained through a realistic simulation environment against those gathered during a real-world flight experiment. The objective is to analyze the controller's effectiveness and robustness in tracking both position and velocity commands, using the Root Mean Square Error (RMSE) and the respective trajectory tracking error plots as the primary quantitative metric to better understand the simulation to real-world gap.

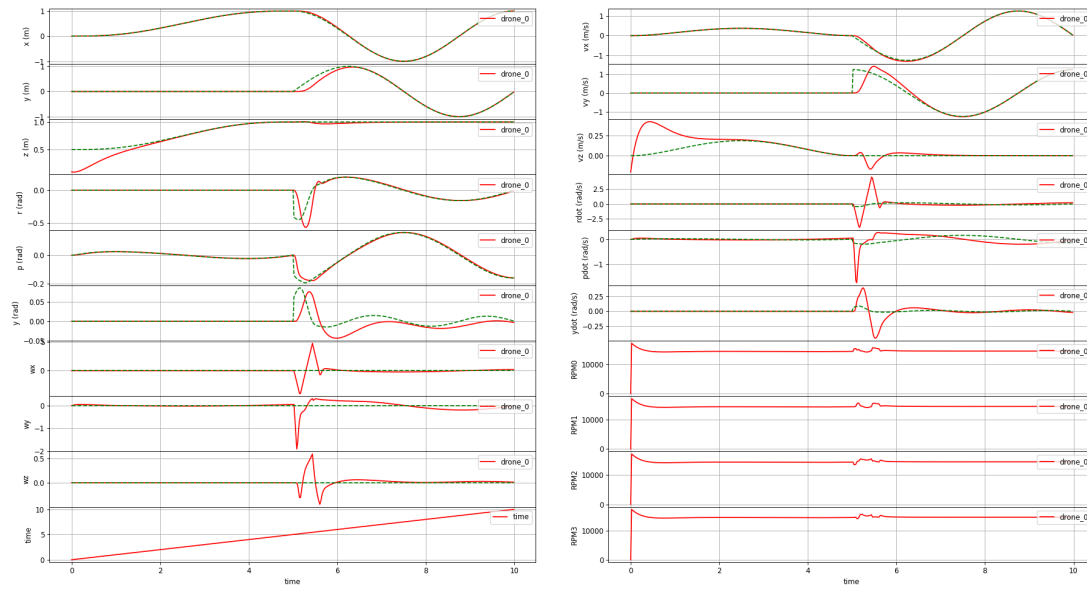
## Baseline Gains

| Controller | Gain Variable | Axes | Tuned Value |
|------------|---------------|------|-------------|
| Position   | Kp            | x, y | 0.16        |
|            |               | z    | 0.20        |
|            | Kd            | x, y | 0.09        |
|            |               | z    | 0.15        |

## Real-world of Circular Trajectory Tracking Error



## Simulation of Circular Trajectory Tracking Error



| Circular Trajectory (RMSE) | Simulation | Real World     |
|----------------------------|------------|----------------|
| RMSE x                     | 0.0130 m   | 0.04127407 m   |
| RMSE y                     | 0.0699 m   | 0.04343791 m   |
| RMSE z                     | 0.0935 m   | 0.28137274 m   |
| RMSE vx                    | 0.7294 m/s | 0.07239376 m/s |
| RMSE vy                    | 0.7054 m/s | 0.06717077 m/s |
| RMSE vz                    | 0.0861 m/s | 0.06214696 m/s |

## 3. Discussion and Conclusion

Overall, the controller demonstrates satisfactory performance in the real-world experiment, successfully tracking the general shape of the circular trajectory without needing to re-tune the parameters. While the controller successfully replicated the trajectory, as shown by the consistently overlaid desired and actual paths, the RMSE analysis shows that the real-world experiment struggled significantly with the altitude control, the z-axis position, yielding an RMSE of 0.2814 m, which is significantly higher than the simulated error of 0.0935 m. This large deviation is clearly visible in the constant offset of the real-world z-position time-series plot. Conversely, the velocity tracking performance in the real-world trials yielded lower RMSE values across all axes compared to the simulation, possibly due to some internal disturbance of the simulated environment. Furthermore, the real-world plots show persistent, high-frequency oscillations in the attitude channels (phi, theta, and psi) not present in the simulation, confirming that the physical controller is continuously battling real-world complexities such as sensor noise, actuator imperfections, and minor air currents that were neglected in the model, ultimately leading to the less accurate position tracking.